

Technical Design - Durations, Dayparts and Bursts Component

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Purpose

This document was originally written for a single component covering Spot Durations, Dayparts and Bursts on the Brief page. That component has since split in two:

1. **Bursts and Custom Dayparts**, staying on the Brief page.
2. **Spot Durations and Standard Dayparts**, moving to the Optimiser Inputs page. Top/Tail and Top/Middle/Tail duration sets move with Spot Durations, since they extend the same selection.

Both components still save their data as Linear Parameter records and follow the same save pattern: their own page-level card, their own shared page save, and their own Fixed-only state.

Custom Dayparts also gains a new capability: a Sales Person can save the current set of custom dayparts as the Account's default, the same way Programs and Program Groups can already be saved as an Account default elsewhere in Capture Brief Details.

Throughout this document, "custom daypart" replaces the earlier "custom time period" terminology. The Linear Parameter Type value was always **Custom Daypart**; only the UI copy and section names are changing to match.

Component Behaviour

Bursts and Custom Dayparts (Brief page)

All trading models

- The user can optionally define one or more burst periods, each with a label, a start date, and an end date.

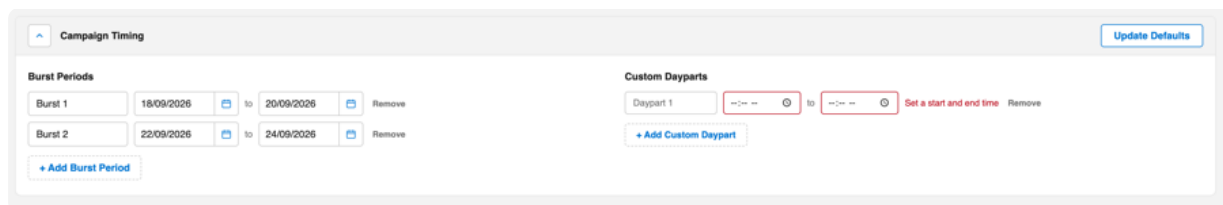


The screenshot shows a UI component titled "Campaign Timing". Inside, there is a section labeled "Burst Periods" which contains a single button labeled "+ Add Burst Period".

Fixed-only trading

If Fixed is the only trading model selected on the brief, one further option appears:

- Add Custom Dayparts.** Each custom daypart is its own row with a label, a start time, and an end time.
- Update Defaults.** Once one or more custom dayparts exist, the user can save the current set as the Account's default.

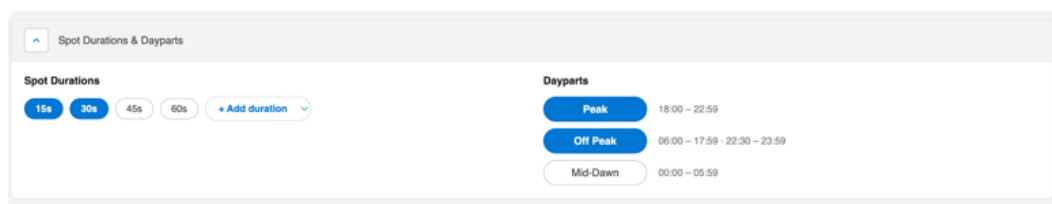


The screenshot shows the "Campaign Timing" UI component with two sections. The "Burst Periods" section on the left contains two rows: "Burst 1" with dates 18/09/2026 to 20/09/2026, and "Burst 2" with dates 22/09/2026 to 24/09/2026. Each row has a "Remove" button and a calendar icon. The "Custom Dayparts" section on the right contains a row for "Daypart 1" with time slots and a "Set a start and end time" button. There is also an "Update Defaults" button in the top right corner and an "+ Add Custom Daypart" button at the bottom of the Custom Dayparts section.

Spot Durations and Standard Dayparts (Optimiser Inputs page)

All trading models

- The user can select multiple spot durations from a predefined list. The four main durations (15s, 30s, 45s, 60s) are presented as pills; further durations are added via a picklist, each addition rendering as its own pill in the same row.
- The user can select one or more standard dayparts from a predefined pill set: Peak, Off-Peak, and Mid-Dawn.



The screenshot shows the "Spot Durations & Dayparts" UI component. The "Spot Durations" section on the left shows four selected pills: "15s", "30s", "45s", and "60s", followed by an "+ Add duration" button. The "Dayparts" section on the right shows three selected pills: "Peak" (18:00 - 22:59), "Off Peak" (06:00 - 17:59, 22:30 - 23:59), and "Mid-Dawn" (00:00 - 05:59).

Fixed-only trading

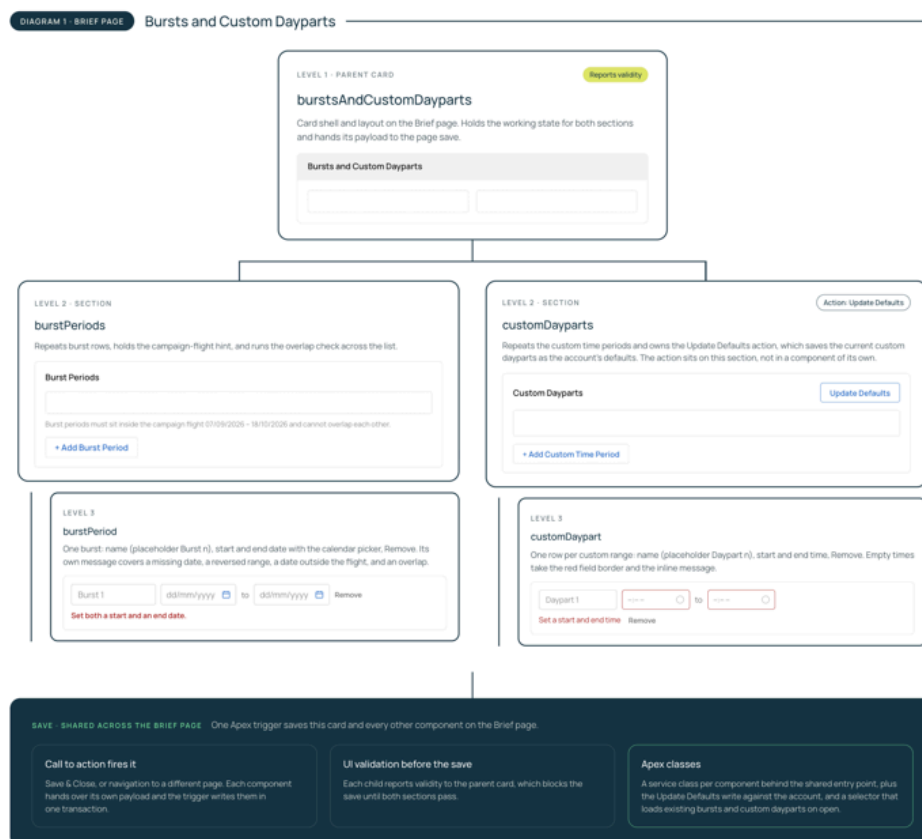
If Fixed is the only trading model selected on the brief, one further option appears:

- **Enable Top/Tail durations.** Once enabled, the user can create one or more Top/Tail duration sets. Each set can optionally take a middle duration, making it a Top/Middle/Tail set.

The screenshot shows a configuration panel titled 'Spot Durations & Dayparts'. It is divided into two main sections: 'Spot Durations' and 'Dayparts'.
Under 'Spot Durations', there is a toggle for 'Enable Top/Tail durations'. Below it are buttons for '15s', '30s', '45s', and '60s', along with an '+ Add duration' button.
Under 'Dayparts', there are three buttons: 'Peak' (18:00 – 22:59), 'Off Peak' (06:00 – 17:59, 22:30 – 23:59), and 'Mid-Dawn' (00:00 – 05:59).

Component Hierarchy

Bursts and Custom Dayparts (Brief page)



Five new Lightning web components across three levels.

Level 1 (parent card)

- `burstsAndCustomDayparts`: the card shell and layout. Holds the working state for both sections and hands its payload to the Brief page's shared save. Reports its own validity upward.

Level 2 (sections)

- **burstPeriods** : repeats burst rows, holds the campaign-flight hint, and runs the overlap check across the list.
- **customDayparts** : repeats custom daypart rows, and now also owns the Update Defaults action, including the check for whether the current set diverges from the stored Account default.

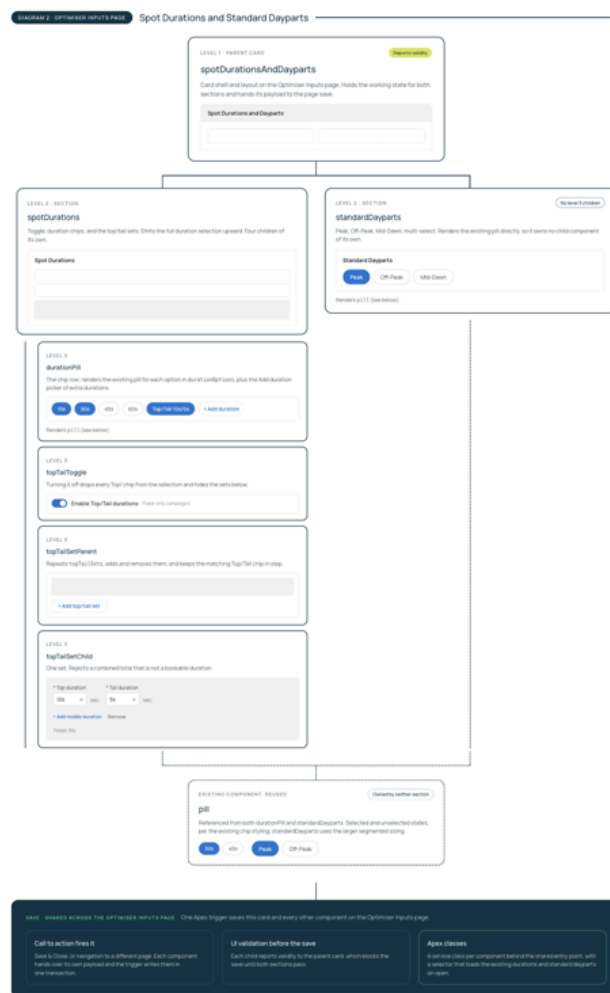
Level 3, burstPeriods children

- **burstPeriod** : one burst, with a name (placeholder "Burst n"), a start date, and an end date via the calendar picker. Its own message covers a missing date, a reversed range, a date outside the campaign flight, and an overlap with another burst.

Level 3, customDayparts children

- **customDaypart** : one row per custom range, with a name (placeholder "Daypart n"), a start time, and an end time. Empty times take the red field border and the inline message.

Spot Durations and Standard Dayparts (Optimiser Inputs page)



Seven new Lightning web components across three levels, plus the existing `pill` component, reused twice.

Level 1 (parent card)

- `spotDurationsAndDayparts` : the card shell and layout. Holds the working state for both sections and hands its payload to the Optimiser Inputs page's shared save. Reports its own validity upward.

Level 2 (sections)

- `spotDurations` : the toggle, duration chips, and the Top/Tail sets. Emits the full duration selection upward.
- `standardDayparts` : standard daypart selection. No longer needs a wrapper shared with custom dayparts, since custom dayparts moved to the other page; renders the `pill` component directly rather than through a Level 3 child.

Level 3, `spotDurations` children

- `durationPill` : the chip row. Renders the existing `pill` component for each option in `durationOptions` , plus the Add duration picker for extra durations.
- `topTailToggle` : the Enable Top/Tail durations toggle. Turning it off drops every Top/Tail chip from the selection and hides the sets below.
- `topTailSetParent` : repeats `topTailSets` , adds and removes them, and keeps the matching Top/Tail chip in step.
- `topTailSetChild` : one Top/Tail (or Top/Middle/Tail) set. Rejects a combined total that isn't a bookable duration.

Existing component, reused

- `pill` : selected and unselected states, per the existing chip styling. Reused inside `durationPill` and `standardDayparts` .

Save Architecture

Each page keeps its own save architecture, run independently of the other.

Bursts and Custom Dayparts (Brief page)

- One Apex trigger saves this card and every other component on the Brief page, not just this one. Each component hands over its own payload; the trigger writes them together in a single transaction, so the components stay unaware of each other.
- Save is triggered by Save & Close, or by navigating to a different page.

- Validation is UI-based and runs when the call to action fires. Each child component reports its own validity to the parent card, which blocks the shared save until both sections pass.
- All derived totals, add/remove logic, and payload shaping live in the component JavaScript controllers.
- Apex is one service class per component behind the shared save entry point, with a selector that loads the existing bursts and custom dayparts on open.

Spot Durations and Standard Dayparts (Optimiser Inputs page)

- One Apex trigger saves this card and every other component on the Optimiser Inputs page, not just this one. Each component hands over its own payload; the trigger writes them together in a single transaction, so the components stay unaware of each other.
- Save is triggered by Save & Close, or by navigating to a different page.
- Validation is UI-based and runs when the call to action fires. Each child component reports its own validity to the parent card, which blocks the shared save until both sections pass.
- All derived totals, add/remove logic, and payload shaping live in the component JavaScript controllers.
- Apex is one service class per component behind the shared save entry point, with a selector that loads the existing durations and dayparts on open.
- This assumes the same single-trigger, save-on-CTA pattern as the Brief page component. The earlier Optimiser Inputs stories (`optimiser-inputs-03` , `optimiser-inputs-04`) described a different pattern, field-level autosave via `OptimiserInputsAutosaveController` .

On a successful save, the push of the updated Linear Parameter records to the OMS happens as part of that same save, not on a later or separate sync.

This matters because the components validate input formats at the JavaScript level before the save is allowed to proceed at all: by the time a save succeeds, every duration, daypart, and burst period on the card is already in a valid, well-formed state. Pushing to the OMS immediately on save means the OMS only ever receives that validated data, rather than holding stale or unconfirmed values until a separate sync catches up.

OMS Save Behaviour

On a successful save of either page, the push of that page's updated Linear Parameter records to the OMS happens as part of that same save, not on a later or separate sync. The two pages push independently of each other, each on its own save.

This matters because the components validate input formats at the JavaScript level before the save is allowed to proceed at all: by the time a save succeeds, every duration, daypart, and burst period on the card is already in a valid, well-formed state. Pushing to the OMS immediately on save means the OMS only ever receives that validated data, rather than holding stale or unconfirmed values until a separate sync catches up.

Data Model

Object and parenting

All data from this component is saved as individual Linear Parameter records related to the Proposal. Each record is created with Proposal ID populated on the Proposal lookup field.

Type picklist values

Five Type values cover this component's records:

1. **Single Duration:** an individual duration, e.g. 10s.
2. **Combo Duration:** a Top/Tail or Top/Middle/Tail duration. The two are distinguished by whether Middle Duration is populated, not by separate Type values.
3. **Burst Period:** a burst period.
4. **Standard Daypart:** Peak, Off-Peak, or Mid-Dawn.
5. **Custom Daypart:** a custom time period.

New fields required

- Duration Value
- Top Duration
- Middle Duration
- Tail Duration
- Label
- Start Date (if not already created for another feature)
- End Date (if not already created for another feature)
- Daypart
- Start Time
- End Time
- External Identifier

Start Date, End Date, Start Time, and End Time are kept generic rather than prefixed per Type (e.g. not "Burst Start Date" or "Daypart Start Time"), so the same fields can be reused by any

future date-bound or time-bound Linear Parameter Type, such as exclusion periods introduced in another story.

6.4 Type and Inclusion Type reference

The Linear Parameter object already carries exclusion-type records from other Capture Brief Details components. The table below sets out every Type value against its Inclusion Type, so the values this component introduces sit consistently alongside the existing exclusion values.

[illegible]

Id	n	d													
record Id	Combo Duration	Included	LEAVE...	LE...	LE...	FR...	FR...	FR...	LE...	LE...	LE...	LE...	LE...	LE...	LE...
record Id	Burst Period	Included	LEAVE...	LE...	LE...	LE...	LE...	LE...	FR...	FR...	FR...	LE...	LE...	LE...	LE...
record Id	Standard Daypart	Included	LEAVE...	LE...	LE...	LE...	LE...	LE...	FR...	LE...	LE...	FR...	LE...	LE...	FR...
record Id	Custom Daypart	Included	LEAVE...	LE...	LE...	LE...	LE...	LE...	FR...	LE...	LE...	LE...	FR...	FR...	LE...

The first five rows are existing exclusion Types from other components on the Brief page; the last five are introduced by this component.

Custom Daypart records now exist in two forms: proposal-scoped records (Proposal ID populated, `AccountId__c` blank) and Account-level default records (`AccountId__c` populated, Proposal ID blank). Both use the same Type value and the same fields.

Custom Daypart Account-Level Defaults

This mirrors the pattern already built for Program and Program Group exclusions, applied to Custom Daypart.

- **Stored default:** a set of Linear Parameter records with Type = Custom Daypart and `AccountId__c` populated to the proposal's Account, Proposal ID left blank. This set represents the Account's current default custom dayparts, independent of any one proposal.
- **Copy forward:** when a new proposal is created for an Account that has a stored default, a service (mirroring the existing `DefaultExclusionService` pattern) copies that Account's default Custom Daypart records onto the new proposal's Linear Parameter records, with Proposal ID populated and `IsApplyByDefault__c` set true on each copied row.
- **Update Defaults button:** appears in the Custom Dayparts section only, and only for Fixed-only trading, since Custom Dayparts itself is Fixed-only. No separate visibility check is needed

beyond that existing gate: there's no scenario where an Account has custom dayparts, and therefore a possible default to update, outside a Fixed-only proposal.

- **Greyed-out logic:** the button stays disabled while every custom daypart currently on the page still carries `IsApplyByDefault__c = true` and the set otherwise matches the stored default exactly (same rows, same label/start time/end time). Adding, removing, or editing any row clears divergence and enables the button, the same as editing a copied-in Program/Program Group exclusion clears its default flag.
- **On click:** the existing Account-level default Custom Daypart records are replaced with a fresh copy of the current on-page set. The next proposal created for this Account then picks up the new default via the copy-forward service.

Data Sources

Spot Durations and Standard Dayparts (Optimiser Inputs page)

- **Durations:** sourced from Agreement Default Ratio records. Get the Agreement Default Ratio records for the proposal's agreement and take the Spot Duration (Seconds) value from each as a picklist option. A duration with no corresponding Agreement Default Ratio record does not render as an option.
- **Standard dayparts:** proposed to create a simple custom object as the source. Save the daypart's external identifier to the Linear Parameter record for ease of lookup.

Bursts and Custom Dayparts (Brief page)

- **Burst periods and custom dayparts:** defined per proposal, with no external data source. Each is a name plus a start and end date or time, configured directly on the component. An Account's stored default custom dayparts come from whatever set of custom dayparts a Sales Person last saved as the default, also with no external source.

Save and Navigation Behaviour

Bursts and Custom Dayparts (Brief page)

Burst periods

- Burst periods must not overlap in dates.
- Burst periods must not fall outside the campaign start and end dates.
- A burst period, once created, must have a label, a start date, and an end date.

Custom dayparts

- Custom dayparts must not overlap.
- A custom daypart, once created, must have a label, a start time, and an end time.

- Custom dayparts are entirely optional. The earlier rule requiring at least one custom daypart when no standard daypart was selected is dropped: standard dayparts now live on a different page's component, so this component no longer checks against them.

Spot Durations and Standard Dayparts (Optimiser Inputs page)

Top/Tail and Top/Middle/Tail durations

- The combined total of the populated fields (Top, Middle, Tail) must equal one of the predefined bookable durations.
- A set, once created, must have a Top Duration and a Tail Duration populated.
- No two sets can duplicate the same combination, e.g. two 10s/5s Top/Tail sets.

Durations and dayparts

- At least one duration must be selected, whether a standard duration or a Top/Tail (or Top/Middle/Tail) set, before the user can proceed beyond the Optimiser Inputs page.
- At least one daypart must be selected, standard dayparts being the only option available on this page, before the user can proceed beyond the Optimiser Inputs page.
- Exact validation trigger point (leaving the page vs. a page-level "Next" action) and message copy are unconfirmed now that this replaces the old pre-navigation gate described below.

Navigation to the Optimiser page

Only required field to navigate to the Optimiser page is the Demo Code selection - otherwise ensurance that all burst periods and custom dayparts don't overlap and are completed is required validation here.

Work Split and Estimated Effort

The components listed here will be developed as part of the following user stories:

- ☒ PROP-1009: [SF] Build Task - Data Model changes to Linear Parameters object (Durations, Dayparts and Bursts) Prioritised
- ☒ PROP-1006: [SF] Build Task - spotDurations In Progress
- ☒ PROP-1007: [SF] Build Task - standardDayParts In Progress
- ☒ PROP-1008: [SF] Build Task - burstPeriods component and Save function Prioritised
- ☒ PROP-964: [SF] Build Task - customDaypartsBursts and save function In Progress

User Stories featuring Component

The following user stories were created to capture this component's functionality:

- ☒ PROP-845: [SF] Select Spot Durations and Dayparts for Optimiser Prioritised

- [🔖 PROP-857: \[SF\] Select top/tail durations for optimiser](#) Prioritised
- [🔖 PROP-947: \[SF\] Define Custom Dayparts for Proposal](#) Ready
- [🔖 PROP-955: \[SF\] Define Bursts for Campaign](#) Prioritised